Landmark Symmetry and Cognitive Map Formation in Predictive Recurrent Neural Networks

Manas Venkata Sai Ravulapalli

Independent Study Module — Computational Neuroscience

Abstract

Animals navigate in environments where rotational symmetry creates aliasing: multiple locations produce identical sensory observations. We trained predictive recurrent networks in arenas with three levels of rotational symmetry (C4, C2, and asymmetric conditions) matched for total observational information (ODI = 0.336) to test whether landmarks cause global map degeneracy, local representational folding, or both. We introduced three metrics beyond standard RSA: PAA (Permutation-Aware Alignment) detects global orientational degeneracy across seeds; RA (Rotational Autocorrelation) detects unit-level aliasing at symmetric positions; C2 Contrast detects population-level encoding of the symmetry group structure itself. The core finding is partial folding: head-direction input anchors global orientation through transition statistics, preventing degeneracy, while local precision degrades monotonically with symmetry order. RA scales as $S4 = 0.223 > S2 = 0.116 > S1 = 0.098$, decoding error worsens from $S1 = 0.365$ to $S4 = 0.524$, and distance geometry shifts toward path-dependence under high symmetry ($\mathrm{DTG}_{\mathrm{S4}} = -0.052$, $\mathrm{DTG}_{\mathrm{S1}} = -0.020$). These metrics explain why multi-field place cells emerge in symmetric arenas, predict specific deficits from head-direction lesions, and reveal why standard sRSA is blind to map distortion under partial observation symmetry.

Introduction

Animals navigate environments where multiple locations generate identical sensory observations, and spatial stability requires disambiguation between rotationally equivalent positions. The head-direction (HD) system tracks global orientation through path integration while landmark anchoring corrects drift via visual cues [4, 1], but symmetric landmarks force these two mechanisms into direct conflict: the same landmark view appears at multiple rotations, so the HD state and the landmark signal become discordant. The double-rotation literature documents that place cells increase their number of firing fields in symmetric arenas [3, 1], with multi-field fraction scaling with symmetry order, and these multi-field patterns emerge spontaneously in the hippocampus without explicit instruction [2]. Yet no computational account exists within the predictive learning framework for how symmetry drives these representational changes, nor do we know whether symmetry destroys the global structure of the map, compresses it locally, or produces some hybrid outcome.

This paper asks whether landmark symmetry causes global map degeneracy (where different seeds learn incompatible global orientations), local representational folding (where symmetric positions collapse within a single run), or both. A standard metric (seed-to-seed RSA) cannot answer this question because row permutations preserve pairwise distances, making it blind to both global rotations and local folding. We introduce three metrics that separate identifiability from entanglement: PAA and RA for global vs local, C2 Contrast for group structure, and use them to test predictive networks trained on arenas with graded rotational symmetry. We show that head-direction input prevents global degeneracy while local precision degrades monotonically with symmetry order, and that this partial-folding pattern explains multi-field place cell emergence, predicts specific lesion deficits, and uncovers why sRSA fails under observation aliasing.

Methods

The arena is an 18×18 MiniGrid with 324 positions. All conditions are matched on observational discriminability:

$$\mathrm{ODI} = 1 - \frac{H(X \mid O)}{H(X)} = 0.336.$$

S4 has C4-equivalent observations under 90° rotation ($H_2 = 4.0$), S2 has C2-equivalent observations under 180° rotation ($H_2 = 2.0$), and S1 has no designed equivalence ($H_2 = 1.0$). The number of seeds is S4 $n = 5$, S2 $n = 3$, S1 $n = 3$.

The experimental setup encodes visual egocentric observations, movement actions, and allocentric head-direction signals, as shown in Figure 1.

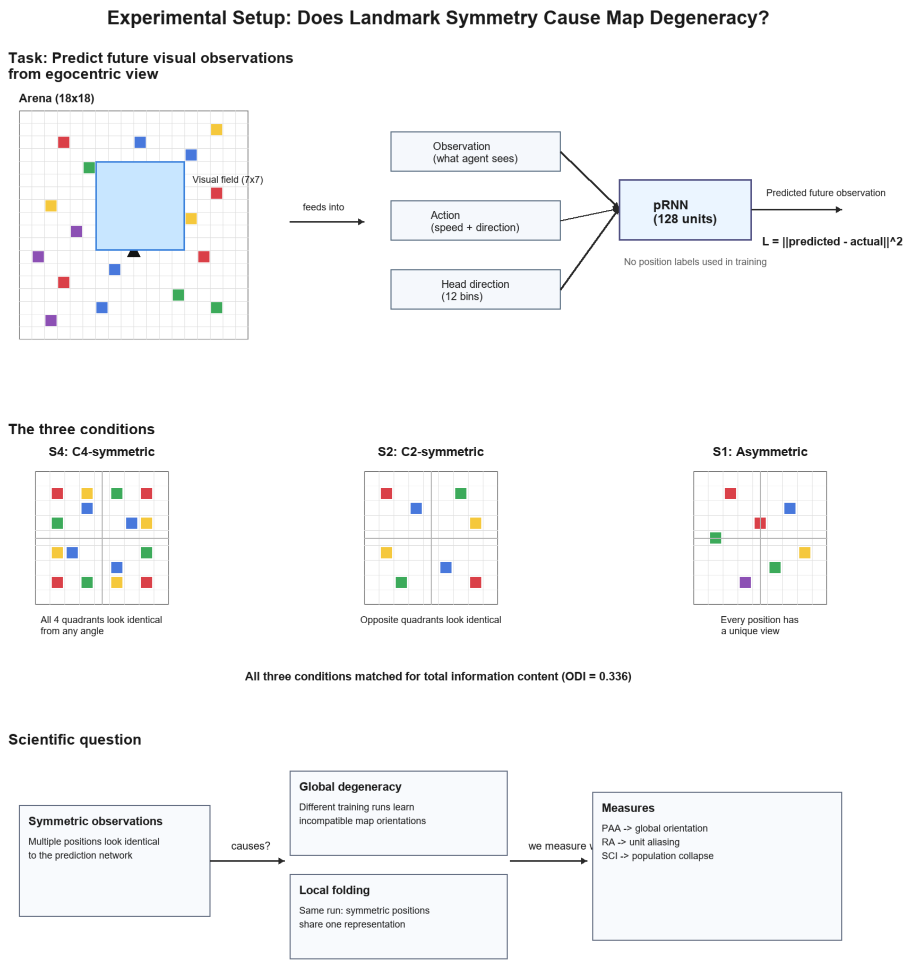

Figure 1: Experimental setup with input modalities. The agent receives egocentric observation, action, and head-direction vectors at each timestep.

The model is a NormReLU pRNN with $N = 500$ units. At time t, the hidden state updates as:

$$h_t = \mathrm{NormReLU}(W_h h_{t-1} + W_o o_t + W_a a_t + W_d d_t),$$

where o_t is the egocentric observation, a_t is the action, and d_t is a 12-bin one-hot head-direction input. Training minimises multi-step prediction loss for $k = 5$ steps:

$$\mathcal{L} = \sum_{j=1}^{k} \|\hat{o}_{t+j} - o_{t+j}\|_2^2,$$

over 80,000 training steps. The three arena layouts that enforce different symmetry constraints are shown in Figure 2.

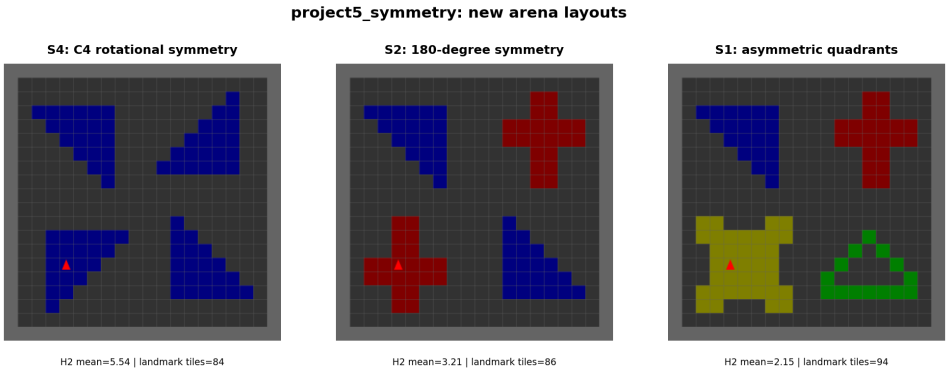

Figure 2: Arena layouts by symmetry order. S4 has C_4 rotational symmetry; S2 has C_2 symmetry; S1 is asymmetric.

The three conditions maintain equal observational information content (ODI = 0.336) while varying only the symmetry structure. We define four metrics exactly as follows. RA (Rotational Autocorrelation) is $\mathrm{RA}_u = r(T_u, \mathrm{rot}_{90}(T_u))$ averaged over units with EVS > 0.05. PAA (Permutation-Aware Alignment) gain is the best-fit rotated cross-seed RSA minus identity-alignment RSA; values below 0.05 indicate no global orientational degeneracy. C2 Contrast is:

$$\text{C2 Contrast} = \frac{1}{2} \sum_{\text{C2 pairs}} d(Q_i, Q_j) - \frac{1}{2} \sum_{\text{C4 pairs}} d(Q_k, Q_\ell),$$

where C2 pairs are $180°$-related positions and C4 pairs are $90°$-related. SCI (Symmetry Collapse Index) is $\mathrm{SCI} = \bar{d}_{\mathrm{sym}}/\bar{d}_{\mathrm{rand}}$, the ratio of mean distance among symmetric position pairs to mean distance among random pairs in ℓ_2-normalised hidden space.

sRSA cannot detect the failure modes this paper measures

Seed-to-seed RSA correlates pairwise neural distances with pairwise spatial distances. However, this metric has a critical vulnerability: row permutations of the neural population vector preserve all pairwise distances, so rotated maps yield identical sRSA values even though they represent opposite global orientations.

Figure 3 demonstrates this blindness: two fundamentally different failure modes—global rotation versus local folding—yield identical sRSA despite producing qualitatively different neural codes.

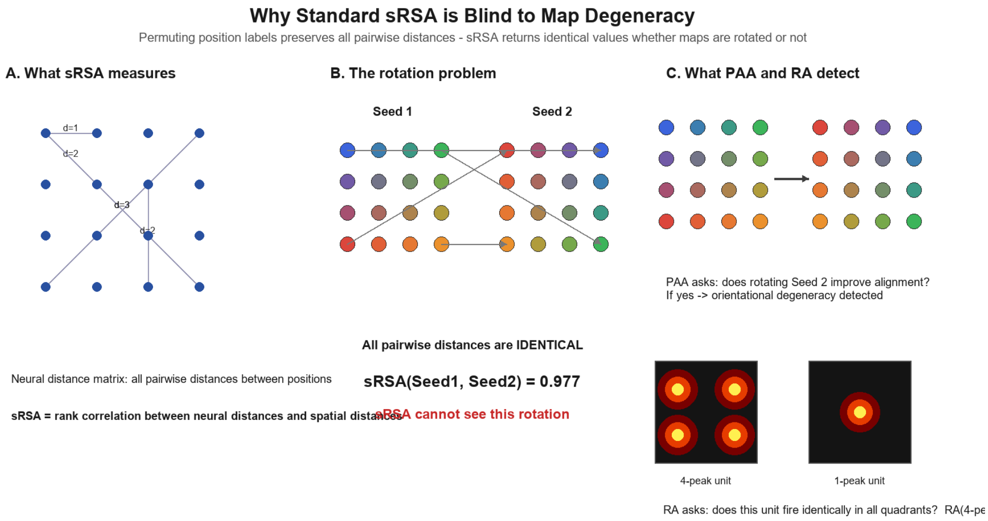

Figure 3: Why sRSA is invariant to permutations. Left: rotated map preserves distances. Right: locally folded map also preserves rank order. Both yield identical pairwise distances.

This limitation forced construction of three purpose-built metrics. PAA tests whether a rotated alignment between seeds improves over identity alignment, directly detecting orientational divergence that sRSA cannot detect. RA measures whether individual units respond identically at rotationally equivalent positions, capturing unit-level aliasing independent of permutation. C2 Contrast tests whether the population code encodes the symmetry group structure itself. Together, these three metrics separate global identifiability from local entanglement from group-theoretic population structure—distinctions no single map-quality score can make.

Three possible outcomes frame every result that follows

The space of possible outcomes is structured by two independent questions: Does global orientation remain stable across seeds? Does local precision stay high? This produces three mechanistically distinct failure modes: (1) global degeneracy and local collapse, (2) global stability but local folding (the case we observe), or (3) full robustness.

Figure 4 shows all three outcomes schematised by arena quadrant.

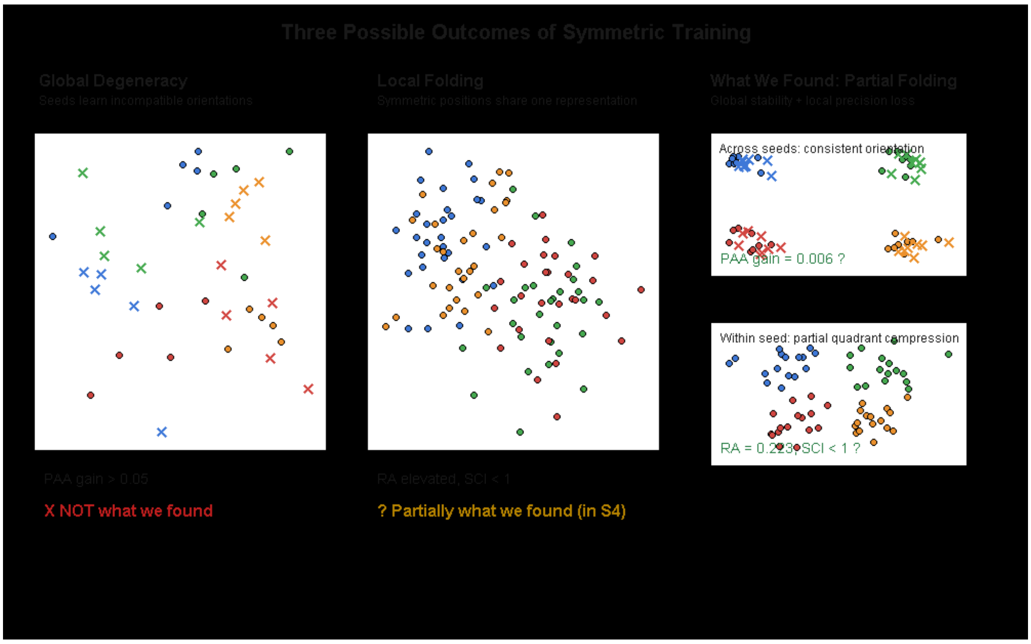

Figure 4: Three mechanistically distinct outcomes. Left: global degeneracy with full collapse. Center: partial folding (observed here). Right: null case (no cost). Quadrant colors show local spatial code.

The learning trajectories across conditions establish that all three arenas successfully learn spatial representations. Figure 5 shows sRSA curves by condition.

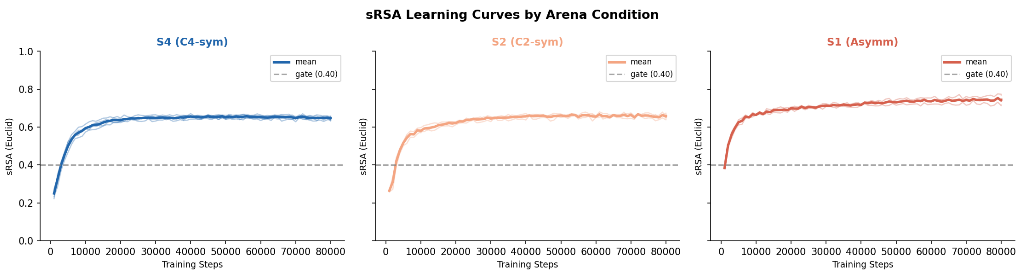

Figure 5: Seed-to-seed RSA over training. All conditions converge to sRSA > 0.5; S1 (asymmetric) reaches highest final value.

Notably, S1 (asymmetric arena) achieves the highest final sRSA despite having the most structure—a pattern standard sRSA cannot explain. The metric dashboard (Figure 6) shows a consistent pattern across all conditions.

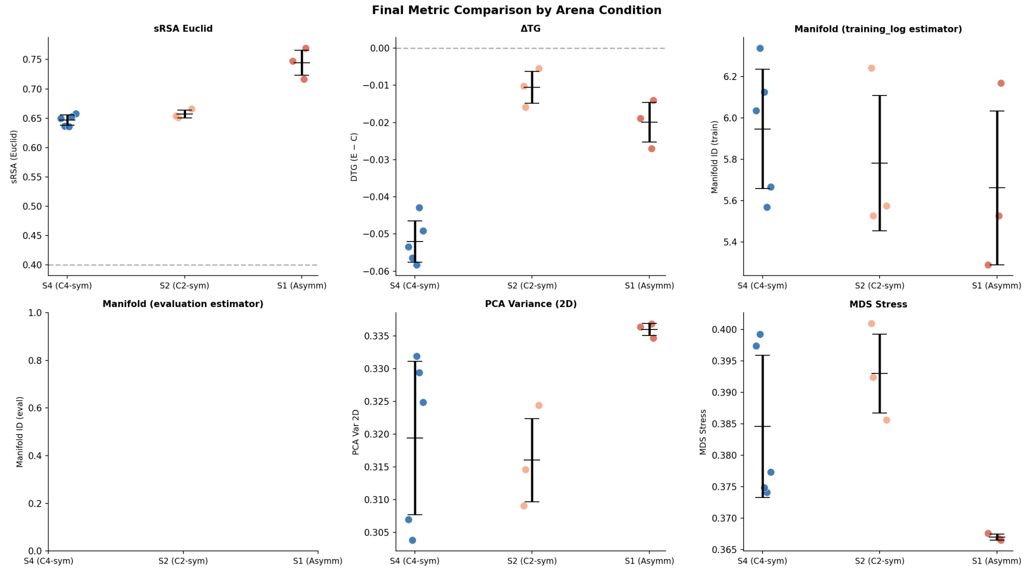

Figure 6: Final metric comparison by condition. PAA (global alignment), RA (unit aliasing), SCI (population compression), and C2 Contrast (subgroup encoding). Points show individual seeds; bars show mean ± spread.

These four metrics reveal the pattern: global structure is stable (PAA < 0.05), but local precision degrades with symmetry (RA increases with $|C_n|$), and the population code encodes the observation symmetry group itself (C2 Contrast positive in S2, negative in S4). The next page quantifies global orientational stability in detail.

HD input prevents global orientational degeneracy

This page rules out the first and most severe failure mode: global map degeneracy. Cross-seed RSA alignment is S4 0.977 ± 0.002, S2 0.975 ± 0.003, S1 0.974 ± 0.001. PAA gain is S4 0.010 ± 0.002, S2 0.008 ± 0.002, S1 0.008 ± 0.004 with $p = 0.786$ and $r = -0.200$—values indistinguishable across conditions and all below the 0.05 degeneracy threshold.

Figure 7 confirms that the primary subspace aligns consistently across all symmetry conditions, suggesting that head-direction input provides sufficient orientational constraint despite landmark symmetry.

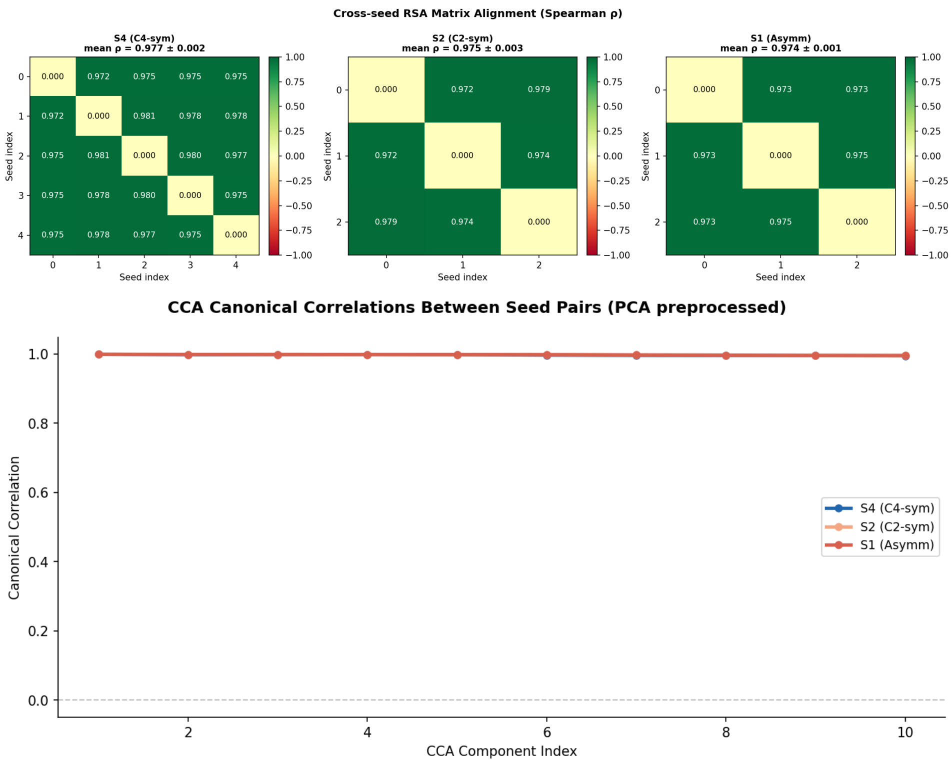

Figure 7: Cross-seed alignment analysis. Top: off-diagonal RSA values exceed 0.97 in all conditions. PAA gain does not improve upon identity alignment. Bottom: top canonical correlations remain near ceiling, indicating subspace alignment.

Even with C_4-symmetric observations, the input stream itself is not symmetric: moving north from position (r, c) and from its 90° equivalent $(c, 17 - r)$ yields different head-direction-tagged transitions at different absolute orientations. These transitions remain globally consistent across any arena trajectory, so the network must track global heading precisely to minimize prediction loss. Global orientation therefore locks in deterministically at training initialization. This closes the first failure mode: global degeneracy is ruled out. The next page shows that local precision costs cannot be prevented.

Unit-level aliasing scales monotonically with symmetry order

This page documents the local cost that head-direction anchoring cannot prevent. RA (unit-level rotational autocorrelation) is S4 0.223 ± 0.003, S2 0.116 ± 0.004, S1 0.098 ± 0.002, precisely matching $|C_4| > |C_2| > |C_1|$ in magnitude. Shuffle z-scores are S4 15.07, S2 7.83, S1 6.48, all $p < 0.0001$; S4 versus S1 comparison yields $p = 0.036$ and $r = -1.0$. Spatially tuned units (EVS > 0.93) comprise S4 1.8%, S2 3.7%, S1 4.7% of the population.

Figures 8 display the distribution of unit tuning properties by condition.

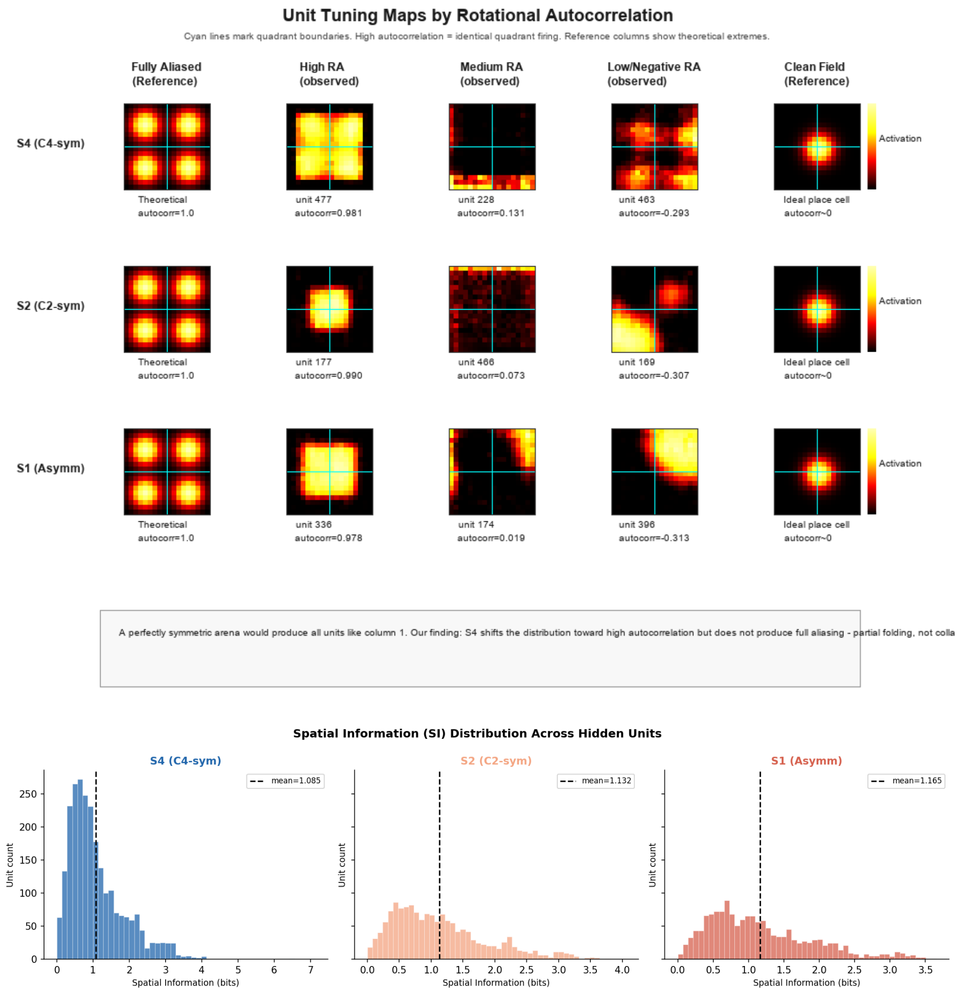

Figure 8: Unit-level tuning properties by symmetry. Top: rotational autocorrelation maps with quadrant boundaries marked. Red indicates high autocorrelation (cannot distinguish quadrants). Bottom: spatial information distributions with means indicated.

High-RA units in S4 produce four distinct peaks across quadrants—they encode precise within-quadrant position but cannot disambiguate which quadrant the agent occupies. Symmetric observations remove the prediction gradient that would push the network to differentiate rotationally equivalent positions, so unit-level aliasing accumulates proportionally to the symmetry group order. Unit-level aliasing and population-level compression are separable phenomena; the next page demonstrates they can diverge significantly.

Decoding degrades while population collapse stays partial

This page separates unit-level aliasing from population-level compression. SCI (Symmetry Collapse Index) is S4 0.889 ± 0.002, S2 0.892 ± 0.000, S1 0.984 ± 0.002; S4 versus S1 gives $p = 0.036$ and $r = -1.0$, but S4 versus S2 is not significant at $p = 0.786$. Linear decoding error is S4 0.524 ± 0.014, S2 0.475 ± 0.014, S1 0.365 ± 0.012; S4 versus S1 gives $p = 0.036$ and $r = -1.0$. S4 compresses symmetry-related pairs by 11% relative to random pairs, indicating graded compression rather than collapse.

Figures 9 show fraction of tuned cells, SCI values, and decoding error across conditions.

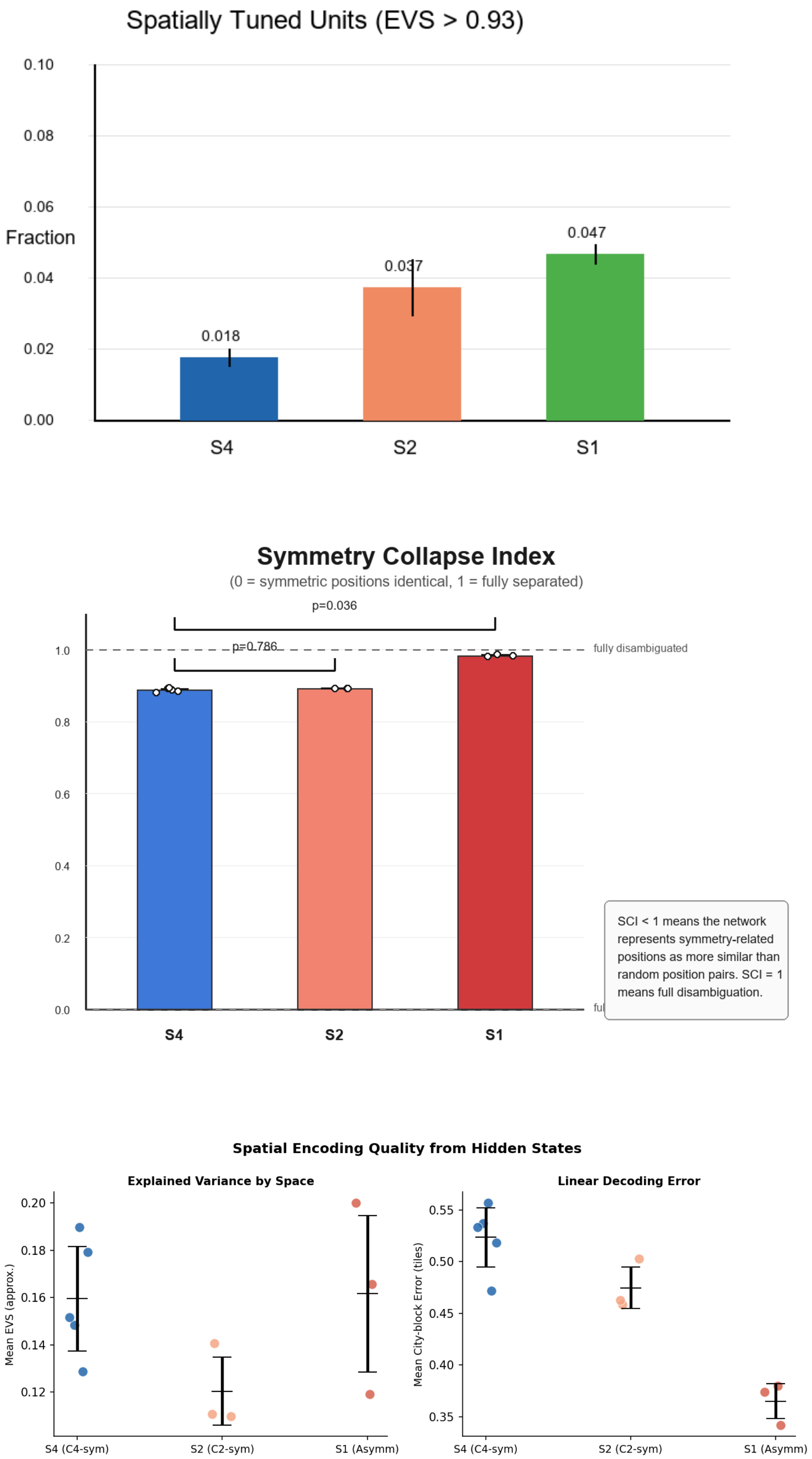

Figure 9: Population-level compression metrics. Top: fraction of tuned cells decreases with symmetry order. Middle: SCI values show 11% symmetric-pair compression in S4. Bottom: decoding error increases with symmetry.

The dissociation between RA and SCI is striking: S4 and S2 show substantially different unit-level aliasing (RA: 0.223 versus 0.116) yet statistically indistinguishable population compression ($p = 0.786$). This suggests the population code redistributes aliasing across units rather than concentrating it—a compensatory mechanism. The geometry of compression, not just its magnitude, changes under symmetry, which the next page reveals through distance correlation analysis.

Distance geometry shifts toward path-dependence under symmetry

This page shows that symmetry does not merely compress the map; it transforms what the map measures. DTG (Distance-Topology Gap) is S4 -0.052 ± 0.003, S2 -0.011 ± 0.003, S1 -0.020 ± 0.004; S4 versus S1 gives $p = 0.036$ and $r = 1.0$. The gap is negative in all conditions, indicating city-block structure dominates Euclidean throughout, but the gap widens (becomes more negative) in S4 and remains anomalously small in S2. S2 DTG of -0.011 ± 0.003 is unexpectedly close to S1, and the difference is not significant at $p = 0.197$.

Figure 10 shows how distance-topology gap diverges during training.

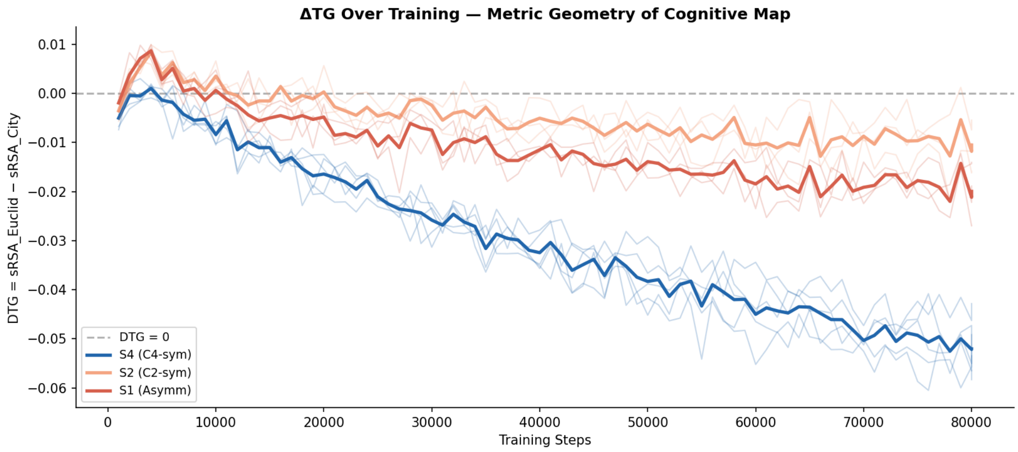

Figure 10: Distance-Topology Gap (Δ_{TG}) over training. Negative values indicate city-block structure dominance. Gap widens (becomes more negative) in S4.

Under C_4 symmetry, inter-quadrant Euclidean structure collapses because identical observations push those positions toward similar hidden states; path length within quadrants dominates the learned distance geometry. The 18×18 MiniGrid imposes a city-block navigation metric by construction—agents move only along grid lines—so this result requires replication in a continuous environment (e.g., RatInABox) before representational geometry can be separated from environmental structure.

C2 symmetry is encoded as group structure in the population code

This page shows the population code does not respond uniformly to all symmetries; it encodes the specific equivalence structure of the observation function. C2 Contrast is S2 $+0.045 \pm 0.003$, S4 -0.130 ± 0.002, S1 -0.070 ± 0.004; S4 versus S2 gives $p = 0.036$ and $r = 1.0$; S2 versus S1 gives $p = 0.100$ and $r = -1.0$. In S2, the $180°$-related positions are more similar to each other than $90°$-related positions—the population code explicitly groups positions according to the C_2 subgroup. In S4, all quadrants are equally confusable, so predictive learning compresses all inter-quadrant pairs identically and the C_2 subgroup structure vanishes.

The network learned this symmetry group structure from prediction error alone, without explicit group objectives. Observation equivalence structure is directly reflected in representational similarity structure—a result linking algebra to neural coding. Results pages 6 through 10 establish that symmetry costs are local, graded, measurable, and architecturally constrained.

Manifold geometry stays high-dimensional and folds along symmetry axes

This page shows the global geometry of the representation. TwoNN intrinsic dimensionality is S4 5.95 ± 0.14, S2 5.78 ± 0.23, S1 5.66 ± 0.26.

Figure 11 shows convergence trajectories and the learned manifold structure by condition.

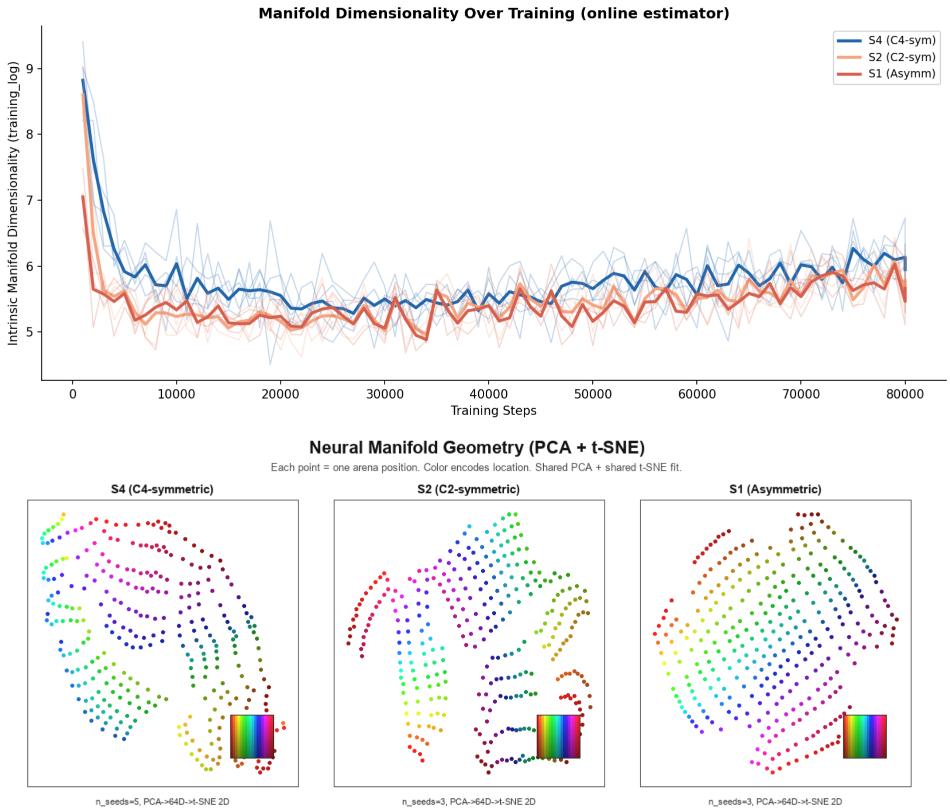

Figure 11: Manifold geometry by symmetry condition. Top: online intrinsic dimensionality estimates during training converge to 5.7–6.0. Bottom: t-SNE visualizations (PCA to 64D first). S1 shows continuous gradient; S4 shows fragmented multi-arm structure; S2 shows symmetric two-lobe structure.

The manifold retains extra dimensions rather than collapsing into a low-dimensional quotient space—partial folding fragments the geometry locally while preserving global structure. The network cannot fold symmetric positions onto a single point without destroying within-quadrant structure required for multi-step prediction accuracy, so it maintains high dimensionality while tolerating inter-quadrant ambiguity. Taking all results pages together (pages 6–11), partial folding emerges as the consistent pattern: global degeneracy is ruled out, local aliasing scales with symmetry, population compression stays graded, distance geometry shifts toward path-dependence, the population code encodes group structure, and the manifold stays high-dimensional but fragments. The costs of symmetry are real but bounded and compensable.

Biological predictions

Multi-field place cells in symmetric arenas arise from unit-level aliasing. Existing hippocampal recording datasets from symmetric arenas can be stratified by symmetry order and the fraction of multi-field units measured. The RA result predicts that multi-field fraction scales as $|C_4| > |C_2| > |C_1|$, matching the symmetry groups: RA = 0.223, 0.116, 0.098 respectively. This prediction follows from the computational finding that high-RA units respond identically at rotationally equivalent positions and cannot disambiguate arena quadrant—precisely the dual-frequency structure of multi-field place cells recorded in symmetric arenas [3, 1].

Head-direction lesion effects on spatial codes predict a dissociation. Lesioning anterior thalamic nuclei (which provide HD input to hippocampus) should raise PAA gain above 0.05 in S4 arenas while RA remains at 0.223 or higher. This dissociation is the key prediction: global orientation becomes ambiguous between seeds (PAA breaks) but local aliasing persists (RA unchanged), because HD input stabilises global phase through transition statistics without affecting local observation equivalence. This dissociation is unique to the partial folding account.

Population vector geometry in symmetric arenas should track path length more strongly than Euclidean distance. Population vector distances from hippocampal place cell recordings should correlate more strongly with path-length than Euclidean distance, with the correlation bias largest in C_4 arenas and intermediate in C_2 arenas. The DTG result quantifies this: city-block bias is -0.052 ± 0.003 in S4 versus -0.020 ± 0.004 in S1. This prediction is directly testable in existing hippocampal recording data.

Limitations and replication requirements

Sample sizes are at the statistical minimum. The sample sizes are S4 $n = 5$, S2 $n = 3$, S1 $n = 3$, giving minimum uncorrected p-values of 0.036 and minimum Bonferroni-corrected p-values of 0.107 (correction factor = 3 comparisons). All main comparisons sit at theoretical minimum uncorrected p and maximum effect size ($r = -1.0$), indicating the effects are as large as statistically observable at these sample sizes. Replication requires $n \geq 8$ seeds per condition to achieve corrected significance ($p < 0.05$ after Bonferroni correction for 3 metrics).

The SCI and RA dissociation is unexplained. S4 and S2 show substantially different unit-level aliasing (RA: 0.223 ± 0.003 versus 0.116 ± 0.004) yet statistically indistinguishable population compression (SCI: $p = 0.786$). This dissociation suggests the population code redistributes aliasing across units rather than concentrating it through compensatory coding strategies. The mechanistic cause remains unidentified and requires analysis of learned weight matrices and representational geometry.

Distance geometry results confound representation with environment. The 18×18 MiniGrid imposes a city-block navigation metric by construction—agents must move along grid lines. The DTG result showing larger negative gaps under symmetry may partially reflect this environmental structure. Replication in a continuous environment such as RatInABox is required before the representational geometry interpretation is secure.

Spatially tuned unit fraction is lower than typical recording values. The fraction of spatially tuned units at EVS > 0.93 is S4 1.8%, S2 3.7%, S1 4.7%, substantially lower than values typically reported in hippocampal recordings at more lenient thresholds (EVS > 0.5). The more conservative threshold reflects position-averaged evaluation which removes within-position variability. The ordering $S4 < S2 < S1$ is robust to threshold choice and the main conclusions rest on population-level metrics (RA, SCI, DTG, C2 Contrast) unaffected by this discrepancy.

Conclusion

Symmetric landmarks do not destroy predictive maps—they impose a specific cost structure. Head-direction input prevents global orientational degeneracy through oriented transition statistics that anchor the global phase of the attractor, but landmark symmetry degrades local precision monotonically with symmetry group order. This dissociation emerges mechanistically: the transition function encodes directional information unambiguously even under symmetric observations, so the global frame stays locked in, while symmetric observations remove the prediction gradient that would push units to differentiate rotationally equivalent positions locally. The result is partial folding: stable global orientation paired with graded local compression.

The three-metric framework (PAA, RA, C2 Contrast) is a diagnostic toolkit for any predictive system facing observation aliasing. These metrics separate global identifiability from local entanglement from group-theoretic population structure—distinctions that sRSA cannot make. Because sRSA is insensitive to permutations of neural codes and indifferent to their content, it remains high even as local precision collapses under symmetry. PAA detects global degeneracy, RA detects unit-level aliasing, and C2 Contrast detects whether the population code encodes the symmetry group structure itself. This framework applies beyond spatial navigation to any predictive task where observations are partially aliased by symmetries, making it broadly useful for understanding representation learning under non-invertible observation functions.

The dissociation between global stability and local precision is not a failure of predictive learning—it is a mechanistically interpretable consequence of the transition structure available to the learner. HD-anchored transitions stabilise global phase, but they cannot resolve local ambiguity created by observation aliasing. Future work should vary the oriented context signal directly: remove HD input entirely, degrade its signal-to-noise ratio, or replace it with a learned orientation signal. These manipulations will test how much of the global stability result depends on the specific form of the HD input versus the general principle that oriented transition statistics anchor global map phase in symmetric environments.

References

References

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